MUNTASIR MAMUN

AI ENGINEER • GENERATIVE AI RESEARCHER • OPEN-SOURCE CONTRIBUTOR

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PROFESSIONAL SUMMARY

AI Engineer and undergraduate researcher specializing in Machine Learning, Generative AI, Large Language Models (LLMs) and Computer Vision. Experienced in building end-to-end RAG pipelines, multi-agent LLM systems, diffusion-based architectures, CI/CD-driven MLOps pipelines and serverless GPU backends. Active open-source contributor to Checkstyle and OWASP with 34+ merged pull requests, bridging deep learning research with scalable production engineering.

TECHNICAL SKILLS

Artificial Intelligence.	: Machine Learning • Deep Learning • Computer Vision • Generative AI • LLMs • AI Agents •RAG • Diffusion Models • Model Fine-Tuning
Frameworks & Libraries	: PyTorch • TensorFlow • Scikit-learn • Hugging Face • LangChain • ComfyUI • OpenCV • NumPy • Pandas • FastAPI • Flask
Cloud & DevOps.	: Docker • Git • GitHub Actions (CI/CD) • AWS • Google Cloud Platform • DigitalOcean • GPU • Infrastructure • Model Deployment
Programming	: Python • C++ • Java • SQL

RESEARCH EXPERIENCE

Undergraduate AI Researcher

Diffusion Architecture Optimization for Virtual Try-On (Ongoing)

- Experimenting with latent diffusion models and garment-alignment pipelines for high-fidelity virtual try-on of traditional apparel.
- Evaluating identity-preservation mechanisms and pose-invariant attention maps, and optimizing reverse-sampling steps to reduce inference latency.

OPEN SOURCE CONTRIBUTIONS

Checkstyle | View on GitHub

- 24+ merged PRs, e.g. migrated XPath regression tests to *verifyXpathWithInlineConfigParser* (#22444, #22367), added compact-source coverage for *ConstantNameCheck* and *NoWhitespaceBeforeCaseDefaultColonCheck* (#22369, #22394), and fixed the site publication date via *project.build.outputTimestamp* (#22342).

OWASP cve-lite-cli | View on GitHub

- 10+ merged PRs passing 8/8 CI checks, e.g. fixed SARIF output to emit *bun.lock* instead of unsupported *bun.lockb* (#1521), added unit tests for finding helpers and for *nearestCommand* / Levenshtein utilities (#1469, #1455), and corrected SPDX and lockfile documentation (#1448, #1454).

SELECTED PROJECTS

VESTON – AI Fashion Virtual Try-On Platform | View on GitHub

Full-stack Generative AI fashion e-commerce application using image-conditioned diffusion models.

- Engineered a virtual try-on pipeline integrating IDM-VTON and Hugging Face diffusers to perform realistic clothing transfer.
- Developed an asynchronous RESTful backend in FastAPI, optimizing request queuing and memory allocation for heavy generative workloads.
- Deployed models on serverless GPU infrastructure, reducing idle operational cost while maintaining low-latency inference.

Technologies: Python, PyTorch, Hugging Face, IDM-VTON, FastAPI, Docker, serverless GPU, GCP, AWS

End-to-End MLOps Pipeline with CI/CD | View on GitHub

Production-style vehicle insurance cross-sell prediction platform with automated training and deployment.

- Structured a 6-stage, config-driven pipeline covering MongoDB ingestion, SMOTEENN imbalance handling and champion-model evaluation, served through a FastAPI web app.
- Containerized the application with Docker and automated testing, image builds and AWS ECR deployment using GitHub Actions CI/CD workflows.

Technologies: Python, Scikit-learn, FastAPI, SMOTEENN, MongoDB Atlas, Docker, GitHub Actions, AWS ECR

Metro Rail AI Assistant | View on GitHub

Multi-modal transit automation system combining speech recognition, route finding and conversational intelligence.

- Built a voice-to-ticket booking pipeline using OpenAI Whisper for high-accuracy automatic speech recognition (ASR).
- Designed a LangChain-driven RAG architecture over vector storage to answer complex, real-time transit queries.
- Implemented intent-classification workflows that parse natural-language queries into accurate route and fare calculations.

Technologies: Python, PyTorch, OpenAI Whisper, MySQL, LangChain, RAG, Vector Databases, FastAPI

Multi-Agent AI Research Assistant | View on GitHub

Autonomous AI agent framework that accelerates academic research and literature analysis.

- Developed a multi-agent orchestration architecture with LangChain for structured research, article retrieval and cross-paper synthesis.
- Engineered prompt-chaining and context-compaction strategies to prevent context-window overflow during multi-document processing.
- Implemented automated fact-verification and citation-tracking modules to reduce LLM hallucinations in generated summaries.

Technologies: Python, LangChain, LLMs, Vector Search, PyTorch, API Integration

EDUCATION & CERTIFICATIONS

Bangladesh University of Professionals (BUP) – B.Sc. in Information and Communication Engineering | Expected Dec 2027 | CGPA: 3.55 / 4.00

Saint Joseph Higher Secondary School – Higher Secondary Certificate (HSC), Completed

Certifications: AI Engineer for Data Scientists, DataCamp (View Certificate) • Data Analyst, DataCamp (View Certificate)